YOSHWAN PATHIPATI

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EDUCATION

Virginia Polytechnic Institute and State University (Virginia Tech) **Aug 2023 - May 2027**
Bachelor of Science, Computer Science Minors: AI, Cybersecurity, and Math *Blacksburg, VA*
- **Achievements:** Dean's List (Top 20% of class)
- **Coursework:** Machine Learning, Computer Systems, Data Structures and Algorithms, Intro to AI, Cloud Software Development, Cryptography

TECHNICAL SKILLS

- **Programming Languages**: Python, Java, C, JavaScript, HTML, CSS, SQL, Ruby
- **Frameworks & Libraries**: React, Node.js, Flask, Ruby on Rails, Express, JAX
- **Databases**: MySQL, MongoDB, PostgreSQL, Redis
- **Developer Tools**: AWS, Azure, GCP, Git, GitHub, GitHub Actions, Docker, Linux/Windows, Nessus, Wireshark, Terraform, VMware, nmap, OpenVAS, Ollama, Llama, Llama3/RAG, pnpm, tfsec
- **Certifications**: AWS Certified Solutions Architect, CompTIA Security+ (in progress)
- **Security & Compliance**: NIST Frameworks, CMMC

RELEVANT EXPERIENCE

Virginia Tech Dept. of Engineering Education **Sep 2026 - Present**
AI Research Assistant - LLM Automation *Blacksburg*
- Develop Python preprocessing scripts and faculty-reviewed prompts for CATME feedback generation through Ollama, ensuring compliance with university data-handling policies and delivering automated feedback to over 100 first-year engineering students
- Advise faculty on model selection and deployment by assessing three model families (Llama, Qwen, and Kimi), their computational requirements, and available Hokie AI and VT ARC resources, leading to the adoption of models that meet performance and cost targets for classroom use
- Collaborate with faculty and another assistant to containerize the CATME feedback pipeline with Docker, standardizing dependencies and setup for a target cohort of 100+ engineering students.

Triple Point Security **May 2026 - Aug 2026**
Cybersecurity Intern *Blacksburg*
- Introduced Amazon EventBridge and Amazon SNS to the compliance team; trained 3 teammates through explanations and documentation, enabling to deploy workflows in AWS Regions.
- Resolved a NIST/CMMC framework mismatch across 4 source publications, establishing the control structure for the team's AWS Security Hub compliance mapping.
- Diagnosed stalled OpenVAS feed synchronization in Docker and bypassed an unresponsive web interface through GMP commands, completing a vulnerability assessment across a 5-host lab.
- Conducted authenticated and unauthenticated Nessus scans across 5 hosts, identifying 20+ critical and 84 high-severity findings; analyzed network traffic through Wireshark packet captures.
- Established an AWS-Azure site-to-site VPN, resolving an IKE policy mismatch and teaching teammates VPN setup; provisioned infrastructure across AWS, Azure, and GCP with Terraform.
- Compared self-hosted Llama 3 to cloud LLMs in vulnerability-report generation using tfsec and Nessus, identified severity-classification errors, missed CVEs, and reliability limits in RAG and document uploads, which led to updated detection rules that reduced false-positive classifications

Hume Center National Security Institute + Aerospace and Ocean Engineering Department **Sep 2025 - Present**
Research Assistant - Full-Stack Development *Blacksburg, VA*
- Led development of SpaceNet, building Flask REST APIs that let five aerospace researchers configure satellite simulations, launch experiments, and monitor telemetry in two concurrent workflows, which streamlined their research process
- Advised aerospace researchers on software capabilities and limitations, translating goals into 4 React dashboards for experiment configuration and live telemetry serving 10+ team members.
- Proposed and deployed a Docker application after assessing hosting cost, reliability, and scaling constraints; automated CI/CD with GitHub Actions, reducing deployment time from 45 to 8 minutes.

Virginia Tech Dept. of Computer Science **Jun 2025 - Aug 2025**
Research Assistant - Backend Development *Blacksburg, VA*
- Refactored 3 Ruby on Rails controllers in OpenDSA's course analytics and student-progress modules, eliminating N+1 queries and reducing database query time by 30%.
- Developed Ruby on Rails backend features for the CS 3114 interactive learning system, collaborating with faculty and contributors to add automated grading and real-time feedback workflows; the enhancements streamlined student-instructor interactions and reduced manual grading effort

PROJECTS

Multimodal Associative Memory via Symmetric Recurrent Networks **Feb 2026 - Present**
- Implement continuous-time dynamics in JAX and a Modern Hopfield core with a Log-Sum-Exp energy function to enable bidirectional pattern completion between audio and visual embeddings.
- Design a symmetric recurrent energy-based model for audiovisual associative memory, applying Equilibrium Propagation to train through local, biologically inspired weight updates.

Organic Farm E-Commerce Platform - Built for family agricultural business **Oct 2025 - Present**
- Built a full-stack e-commerce platform with React, Node.js, Express, and MongoDB for the family agricultural business, supporting product listings, inventory management, and order processing.
- Integrated Stripe payments and inventory tracking designed to manage 100+ organic produce SKUs and replace spreadsheet-based order processing.

CloudGuard - Cloud Security Posture Monitoring **Jan 2026 - Apr 2026**
- Provisioned Amazon GuardDuty, Amazon Inspector, and AWS Security Hub with Terraform; automated mapping of Security Hub findings to NIST SP 800-53 controls for cloud security review.
- Integrated tfsec into the CI pipeline to detect infrastructure misconfigurations and block merges when security checks fail.